



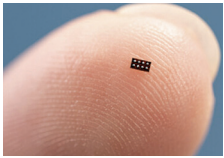
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

The world's smallest MCU

Texas Instruments (TI) has introduced the world's smallest microcontroller (MCU), designed to optimise size and performance for compact applications like medical wearables and personal electronics. The MCU is 38% smaller than the industry's smallest existing MCU, allowing designers to reduce board space without sacrificing performance.



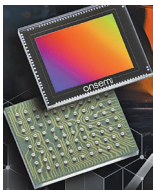
The MSPM0C1104, in a wafer chip-scale package (WCSP), measures just 1.38mm²—about the size of a black pepper flake. Consumers expect more features in smaller and more affordable products, from electric toothbrushes to stylus pens. This compact MCU integrates high-speed analogue components, allowing engineers to maintain computing performance without increasing board size.

Texas Instruments

<https://www.ti.com>

The industry's first real-time iToF sensor

Onsemi has introduced the Hyperlux ID family, the industry's first indirect time-



of-flight (iToF) sensor for depth measurements and 3D imaging of moving objects.

Using global shutter pixel architecture and on-board storage, these sensors capture scenes while processing depth data in real time without external memory or processors. This enables depth sensing up to 30 metres—four times farther than standard iToF sensors—within a small design. The

sensors generate both monochrome images and depth data, removing the need for separate visual and depth sensors. The Hyperlux ID family supports automation, robotics, manufacturing, logistics, agriculture, and access control systems by improving navigation, defect detection, and quality control.

Onsemi

<https://www.onsemi.com>

Powerful MCUs for smart, multi-use AI

Synaptics Incorporated's SR series is a set of microcontroller units (MCUs) for edge AI. The series includes three tiers—performance (100GOPS), efficiency, and ultra-low-power always-on—for scalable, context-aware AI processing.



It integrates accelerators and vision, audio, and voice algorithms while keeping a small form factor. The MCUs include camera inter-

faces and peripherals to reduce system cost, power use, and footprint, making them suitable for security cameras, sensors, appliances, point-of-sale systems, digital signage, and scanners. The series is suitable for IoT device makers, robotics engineers, consumer electronics companies, industrial equipment makers, security system developers, and embedded system engineers.

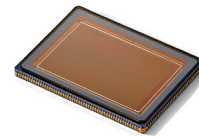
Synaptics Incorporated

<https://www.synaptics.com>

World's first 410-megapixel CMOS 35mm full-frame sensor

Canon has developed the world's first 410-megapixel CMOS 35mm full-frame sensor, designed for applications requiring extreme resolution. With a

resolution equivalent to 24K—198 times Full HD and 12 times 8K—it allows significant cropping and enlargement while preserving detail. Unlike other



high-pixel-count sensors, which are usually medium format or larger, this sensor fits

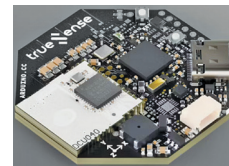
within a 35mm full-frame format, making it compatible with existing full-frame lenses. The sensor is designed for surveillance agencies needing high-detail monitoring, medical professionals for advanced imaging, industrial teams for precision inspections, filmmakers and photographers for ultra-high-resolution content, and researchers in fields like astronomy and microscopy where capturing fine details is essential.

Canon

<https://global.canon/en>

Real-time tracking and gesture control with UWB

Arduino Stella, designed to work with the Portenta UWB Shield, enhances IoT projects using UWB technology for accurate tracking and seamless integration. Built with the nRF52840



microcontroller and Truesense DCU040

module, Stella enables real-time tracking,

automated safety systems, and intuitive human-machine interactions. It supports precise location tracking, secure communication, and easy development with Arduino tools. Applications include warehouse automation with Follow-Me AGVs, secure transportation monitoring, hands-free access control, EV charging automation, and asset